



Helicobacter pylori and Peptic Ulcer

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

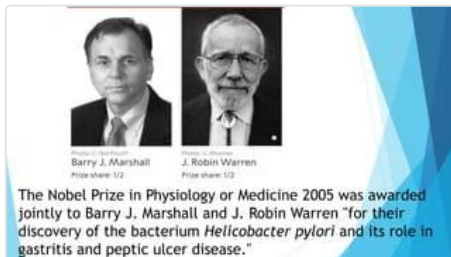
1. How does *H. pylori* survive the acidic gastric lumen (pH 1–3) when it grows optimally at pH 6–7?
2. Which two virulence factors (CagA and VacA) are most directly linked to gastric ulcer/carcinoma risk, and how do they act?
3. Why is serology unsuitable for confirming eradication after treatment, and what tests should be used instead?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

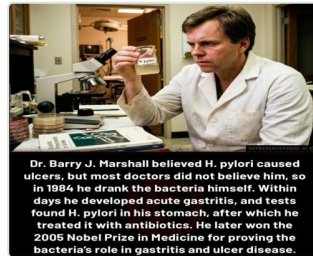
1 Peptic ulcer disease — definition and historical turning point

- A **peptic ulcer** = mucosal defect in the **stomach or duodenum** produced in an acid-peptic environment
- Two major causes dominate clinical practice: ***H. pylori* infection** and **NSAIDs**
- *H. pylori*-associated ulcers are an **infectious disease problem** — eradicating the organism changes recurrence risk

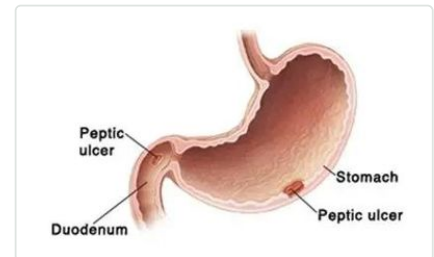
1982: spiral bacteria associated with gastritis were recognized as a major cause of gastric disease, changing peptic ulcer from an "acid-only" disease to one with a treatable bacterial cause. **Marshall and Warren** received the **2005 Nobel Prize** for this discovery; Marshall famously drank a culture of *H. pylori* himself, developed acute gastritis, and proved Koch's postulates for the organism.



Marshall and Warren — 2005 Nobel Prize in Physiology or Medicine.



Marshall's self-experiment: he drank the bacteria, developed gastritis, and later won the Nobel Prize.



Peptic ulcer sites — stomach and duodenum.

H. pylori is linked to	Detail
Chronic active gastritis	Universal finding with colonization
Duodenal and gastric peptic ulcer disease	85% of gastric ulcers, 95% of duodenal ulcers
Gastric adenocarcinoma	~100-fold increased risk via chronic gastritis → atrophy → metaplasia
Gastric MALT lymphoma	From chronic lymphoid infiltration of gastric mucosa

2 Morphology, growth requirements, and transmission

Feature	Detail
Morphology	Curved, helical/spiral, Gram-negative rod ; highly motile via multiple polar flagella (corkscrew motility); older cultures may show coccoid forms
Metabolism	Microaerophilic ; strong urease producer ; usually oxidase-positive and catalase-positive
Culture	Requires enriched media (blood, serum, charcoal); humid microaerophilic incubation at 35–37°C ; colonies may take several days — plates may need prolonged incubation before reporting negative
Culture sensitivity decreases with	Delayed transport, prior antibiotics, suboptimal specimen handling

- Occurs **worldwide**; the **human gastric mucosa is the major habitat**
- **Person-to-person spread** — **fecal-oral** and **oral-oral** routes are important
- Untreated colonization can **persist for decades or for life**

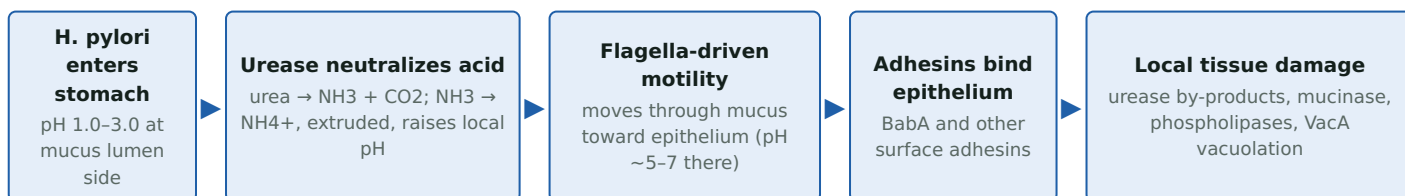
- **Most colonized people never develop ulcer or cancer** — outcome depends on bacterial, host, and environmental factors



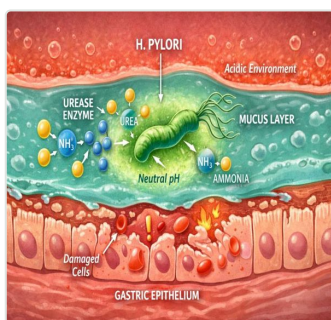
H. pylori — curved, spiral, flagellated Gram-negative rod.

3 Virulence factors and pathogenesis

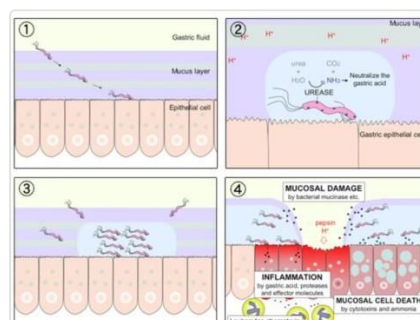
Factor	Role
Urease	Acid survival and tissue irritation
Flagella	Mucus penetration and colonization
Adhesins (e.g., BabA)	Epithelial attachment
VacA (vacuolating cytotoxin A)	Vacuolation and epithelial dysfunction ; also activates IL-8-mediated acute inflammation
CagA (cytotoxin-associated gene A)	Altered host cell signaling and inflammation
Mucinase, phospholipases, inflammatory mediators	Mucosal injury



Gastric mucus is relatively impermeable to acid with strong buffering capacity, creating the pH gradient *H. pylori* exploits.



Urease-mediated acid neutralization allows *H. pylori* to cross the mucus layer toward the epithelium.



Stepwise mucosal disease progression driven by persistent colonization and inflammation.

CagA is secreted and translocated into epithelial cells via a **type IV secretion system**; it disrupts the cytoskeleton and induces **IL-8 → neutrophil attraction**. **CagA-positive strains are more virulent** and directly linked to acute gastritis, gastric ulcer, and gastric carcinoma. **VacA** affects the cell death/proliferation balance and also drives IL-8-mediated inflammation.

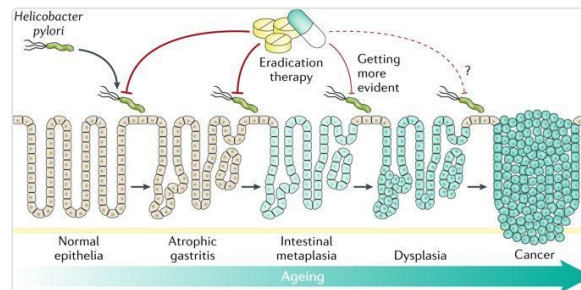
4 Clinical diseases and presentation

Phase	Features
Acute gastritis	Feeling of fullness, nausea, vomiting, hypochlorhydria

Chronic gastritis	Develops if acid secretion stays suppressed; 10-15% progress to peptic ulcer , at sites of intense inflammation — corpus/antrum junction (gastric ulcer) or proximal duodenum (duodenal ulcer)
Gastric cancer pathway	Chronic gastritis → fibrosis + intestinal-type epithelium replacement → ~100-fold increased cancer risk ; higher with CagA-positive strains and high IL-8
MALT lymphoma	Lymphoid infiltration → monoclonal B-cell population in a small subset of patients



The long-term pathway from infection to cancer (Correa cascade).



Progression from normal epithelium through atrophy and metaplasia to gastric cancer.

Presentation	Detail
Epigastric burning / dyspepsia	Nausea, bloating
Duodenal ulcer pain	May occur when the stomach is empty or at night ; improves temporarily with food
Gastric ulcer discomfort	May worsen after meals
Note	Symptoms alone cannot reliably localize the ulcer — endoscopy is definitive ; frequently accompanied by GI bleeding ; no bacteremia or disseminated disease occurs

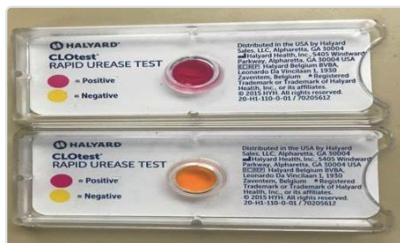


Endoscopic appearance: duodenal ulcer (DU) vs gastric ulcer (GU).

5 Laboratory diagnosis

Invasive tests (require endoscopy)	Non-invasive tests
Histology of gastric biopsy	¹³C Urea breath test
Rapid urease test on biopsy	Stool antigen test
Culture with susceptibility testing	Serology — shows exposure only, not reliable for confirming cure
Biopsy-based molecular testing (where available)	

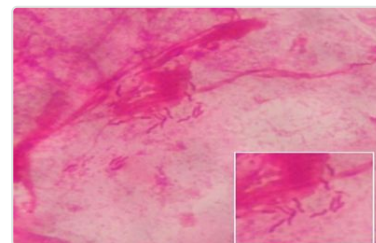
- **False-negative causes:** recent **PPI use**, recent **antibiotics**, low organism density/patchy colonization, inadequate biopsy sampling or improper transport for culture
- **Key limitations:** biopsy tests require endoscopy; culture is slow and technically demanding; **serology cannot prove active infection or cure**



Rapid urease test (CLOtest) — color change indicates a positive result.



Non-invasive urea breath test.



Histology of gastric biopsy (H&E).

6 Treatment and test of cure

Detail	
Standard regimen	PPI (e.g., omeprazole) + clarithromycin + amoxicillin + metronidazole, for 7–10 days
Treatment failure	Most commonly due to clarithromycin resistance ; perform susceptibility testing if the patient fails to respond
Test of cure	Never use symptom improvement as proof of eradication ; use urea breath test, stool antigen test, or a biopsy-based test
Timing	Perform test of cure ≥4 weeks after completing therapy ; withhold treatment ~2 weeks before testing (when appropriate) to reduce false negatives
Standard of care	Every treated patient should have eradication confirmed

P Must-remember pearls

- *H. pylori* = **curved, spiral, Gram-negative, microaerophilic rod**, strongly **urease-positive**, oxidase- and catalase-positive; corkscrew motility via polar flagella.
- **Urease** → neutralizes local acid (survival); **flagella** → penetrate mucus; **BabA** → adhesion; **VacA** → vacuolation/IL-8; **CagA** → type IV secretion, cytoskeletal disruption, IL-8, **most virulent strains**.
- **85% of gastric ulcers, 95% of duodenal ulcers** are *H. pylori*-associated; **10–15%** of chronic gastritis progresses to peptic ulcer.
- Diagnosis: **invasive** (biopsy histology/rapid urease/culture) needs endoscopy; **non-invasive** (urea breath test, stool antigen) confirm **active infection**; **serology only shows past exposure — never use it to confirm cure**.
- Treatment = **PPI + clarithromycin + amoxicillin + metronidazole × 7–10 days**; **clarithromycin resistance** is the top cause of failure; confirm eradication **≥4 weeks post-therapy** with a non-serologic test.

Q Test yourself

1. Which bacterial property is most directly responsible for local neutralization of gastric acid around *H. pylori*?

- A. Urease activity
B. Mucinase activity
C. Oxidase activity
D. Catalase activity

2. Which test is least appropriate for confirming successful eradication of *H. pylori* after therapy?

- A. Serum IgG antibody (serology)
B. ¹³C urea breath test
C. Stool antigen test
D. Biopsy-based test

3. A 30-year-old man has mid-epigastric pain that wakes him at night and improves immediately after meals. A urea breath test is positive. Which description correctly characterizes the causal organism?

- A. Gram-negative curved rod; microaerophilic
B. Gram-negative rod; aerobic
C. Gram-negative rod; facultative anaerobe
D. Gram-positive rod; aerobic

4. Which virulence factor is secreted via a type IV secretion system and directly associated with acute gastritis, gastric ulcer, and gastric carcinoma?

- A. CagA
B. Urease
C. BabA
D. Mucnase

5. Why should a test of cure not be performed immediately after finishing H. pylori eradication therapy?

- A. Residual organisms may transiently suppress test positivity, risking a false negative if tested too early
B. The organism takes 4 weeks to die after antibiotics are started
C. PPIs must be continued for 4 weeks before testing can begin
D. Symptom relief itself is diagnostic and no further testing is needed

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A — Urease breaks down urea into ammonia and CO₂, neutralizing local gastric acid and allowing survival.

2 — A — Serology only shows past exposure/antibody levels and cannot reliably confirm active infection or cure.

3 — A — H. pylori is a Gram-negative, curved/spiral, microaerophilic rod — consistent with classic duodenal ulcer symptoms and a positive urea breath test.

4 — A — CagA is translocated into epithelial cells via a type IV secretion system, disrupts the cytoskeleton, drives IL-8-mediated inflammation, and is linked to the most severe outcomes.

5 — A — Testing too soon after therapy risks false negatives; test of cure should be performed at least 4 weeks after completing therapy, with treatment withheld beforehand when appropriate.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

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Study alongside the lecture, not instead of it.